

Facts for 11 and 12; Remainders

Mission

Warm-up and worked example

Name: _____

Date: _____

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YOUR MISSION Derive 11s and 12s and interpret remainders in context.

SMART MOVE Turn division into a missing multiplication fact.

Math Talk

How can 10×8 help with 11×8 and 12×8 ?

My idea:

New Words

factor	product	quotient
My meaning or picture:	My meaning or picture:	My meaning or picture:

Worked Example

Do this example with your teacher. Say what changes and what stays the same.

1. Derive 11s as 10 groups plus 1 group.
2. Derive 12s as 10 groups plus 2 groups.

What the example shows: _____

Facts for 11 and 12; Remainders

Mission

Learn it and show your thinking

Name: _____

Date: _____

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YOUR MISSION Follow the learning path, then explain the big idea.

SMART MOVE Draw before asking for a rule.

[] **Step 1:** Derive 11s as 10 groups plus 1 group.

[] **Step 2:** Derive 12s as 10 groups plus 2 groups.

[] **Step 3:** Divide 43 counters into groups of 6.

[] **Step 4:** Record $43 \div 6 = 7$ remainder 1.

[] **Step 5:** Discuss whether a context requires 7 groups, 8 groups, or a leftover.

My model, drawing, or organized work:

The big idea in my own words:

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Practice lab

Name: _____

Date: _____

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YOUR MISSION Use a model, equation, or explanation for every task.

SMART MOVE Check whether your answer is reasonable.

1. Mixed 11 and 12 facts.

2. $29 \div 4$, $38 \div 6$, and $50 \div 8$ with models.

3. Explain what each remainder means in a story.

Choose one answer and prove it another way:

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Challenge and final check

Name: _____

Date: _____

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YOUR MISSION Try an unfamiliar problem and defend your reasoning.

SMART MOVE A wrong first idea can still lead to a strong solution.

Challenge Mission

There are 43 students and each table seats 6. How many tables are needed? Explain why the answer is not simply 7 remainder 1.

My plan:

My solution and proof:

Final Check

Write one context in which a remainder is discarded and one in which it causes rounding up.

☐ I can teach it

☐ I need practice

☐ I need help

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Mission

Reusable math tool

Name: _____

Date: _____

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YOUR MISSION Use this page to draw, model, organize, and check.

SMART MOVE Turn division into a missing multiplication fact.

Model and Record

Model	Equation	Check	Explain